	Mean	Std. Dev.	Median	Min.	25th Pctile	75th Pctile	Max.
SSA	0.011	0.024	0.014	−0.056	0.000	0.029	0.040
LATAM	0.021	0.018	0.028	−0.023	0.011	0.031	0.044
SA-APAC	0.019	0.010	0.021	−0.001	0.014	0.027	0.035

Table A.2: Technology advancement factor g, summary statistics (1990-2020).

	Mean	Std. Dev.	Median	Min.	25th Pctile	75th Pctile	Max.
SSA	0.208	0.250	0.111	0.003	0.027	0.305	0.890
LATAM	0.087	0.265	0.045	0.001	0.011	0.120	0.483
SA-APAC	0.115	0.152	0.064	0.003	0.009	0.164	0.481

Table A.3: Increase in pollution stock per unit output k_e, summary statistics (2013-2023).

statistical significance, we prefer to adopt a two-step estimation strategy.

First, we calibrate the effect of technological advancement on the relationship between output and pollution in (6). As there is no measure of physical pollution, we use CO_2 emissions as a proxy. This information is gathered from The World Bank that reports CO_2 emissions (kg per PPP$ of GDP) for a panel of 229 countries[5]. The dataset spans from 1990 to 2020. To estimate the rate of technological progress (g) in the pollution stock process, we fit the time series of CO_2 emission intensity (emissions divided by GDP) to an exponential decay model. Availability of information narrows the clusters to 31 countries in the SSA region, 27 countries in the LATAM region, and 25 countries in the SA-APAC region. In Table 2, 3, and 4 we provide the estimate of g for each of the three macro-regions (median values across the sample of the parameters). Summary statistics of the estimated parameters across countries in each macro-region are shown in Table A.2.

Given the estimate of g, we estimate the remaining parameters (k_e, φ, σ_f) using a ML estimator at the macro-region level. Results (median values across the sample of the parameters) are provided in the second column of Table 2, 3, and 4 for each of the three macro-regions. Summary statistics of the estimated parameters across countries in each macro-region are shown in Table A.3, A.4, A.5, respectively.

A.2 Endowment process

To estimate the endowment process (1), we start with the analysis of natural disaster frequency using climate-related event data from 1980 to 2022, collected by the IMF[6].

[5] `https://data.worldbank.org/indicator/EN.ATM.CO2E.PP.GD`.
[6] `https://climatedata.imf.org/search?collection=Dataset`.

	Mean	Std. Dev.	Median	Min.	25th Pctile	75th Pctile	Max.
SSA	0.38	0.27	0.31	0.04	0.19	0.54	0.85
LATAM	0.40	0.27	0.34	0.02	0.27	0.52	0.81
SA-APAC	0.40	0.22	0.35	0.07	0.25	0.61	0.72

Table A.4: Naturals sink rate φ, summary statistics (2013-2023).

	Mean	Std. Dev.	Median	Min.	25th Pctile	75th Pctile	Max.
SSA	0.029	0.019	0.025	0.002	0.013	0.039	0.065
LATAM	0.030	0.014	0.028	0.010	0.019	0.042	0.053
SA-APAC	0.022	0.017	0.022	0.004	0.009	0.027	0.058

Table A.5: Volatility of the pollution stock process σ_f, summary statistics (2013-2023).

These data come from the Emergency Events Database (EM-DAT) which is maintained by the Centre for Research on the Epidemiology of Disasters (CRED) at the Université Catholique de Louvain[7]. To focus on pollution-related events, we only include those related to floods, extreme temperatures, storms, mass movements, wildfires, droughts, and glacial lake outburst/floods. CRED's database covers a vast number of events ($7,598$ across 229 countries), but many records lack the detailed information to evaluate the severity of the event. In Table A.6 we report some descriptive statistics on the dataset.

The frequency of natural events is notably higher than expected for rare disasters. In our setting, the endowment process (1) aims to capture significant declines associated with rare natural disasters through the jump process and frequent, smaller reductions (somehow structural) through the damage function. Therefore, we have to distinguish rare, severe climate disasters from more frequent minor events. To this end, we extract $\{k(t_i)\}_{i=1}^{10}$ from the CRED database, where $k(t_i)$ represents the number of climate-related events in the *i-th* year for a selected country in scope. We then discretize the endowment process with yearly time steps ($\Delta t = 1$) as follows:

$$\Delta \ln y(t_{i+1}) = \ln\left(y(t_{i+1})\right) - \ln\left(y(t_i)\right) = \varepsilon(t_{i+1}) - \eta(t_{i+1}) \tag{33}$$

where, $\varepsilon(t_{i+1})$ represents the part of the endowment process (1) (white noise component) unaffected by climate risk, drawn from a Normal distribution with mean $\mu - \sigma_y^2/2$ and variance σ_y^2. The impact of climate risk is captured by the term $-\eta(t_{i+1})$. We assume that each climate-related event in the CRED data is independent of the others, with recovery following an exponential distribution with a country-specific scale parameter λ_f. Through

[7] `https://doc.emdat.be/docs/data-structure-and-content/emdat-public-table/`.

	Mean	Std. Dev.	Median	Min.	25th Pctile	75th Pctile	Max.
SSA	0.84	0.56	0.70	0.05	0.37	1.28	2.35
LATAM	1.38	1.28	1.09	0.09	0.37	1.85	4.93
SA-APAC	2.84	4.25	0.93	0.09	0.23	3.62	17.16

Table A.6: Natural events' count per country per year, summary statistics (1980-2022).

some algebra, we find that:

$$\eta(t_i) = 1 - e^{-\xi(k(t_i),\lambda_f)} \tag{34}$$

where $\xi(k(t_i), \lambda_f)$ is sampled from an Erlang-k distribution with event count $k(t_i)$ and scale parameter λ_f. We estimate the parameters μ, σ_y and λ_f for each country in the sample from IMF output data using a ML estimator under the assumption that $\eta(t_i)$ is distributed as in (34). Table A.7, A.8, and A.9 shows summary statistics for the parameters across the three macro-regions, with median values for μ and σ_y detailed in Table 2, 3, and 4. λ_f represents the average annual log-output reduction per climate-related event, with median values of 0.27% in the SSA, 0.31% in the LATAM, and 0.27% in the SA-APAC macro-region.

The above estimates are obtained considering all the climate events, we want now to disentangle rare/severe disasters from frequent/minor disasters. According to (1), the impact of climate risk on the log-output $\eta(t_{i+1})$ depends on pollution stock and is given by:

$$\eta(t_{i+1}) = \mu D(f(t_i)) + \big(1 - Z(t_i)\big)N(t_i) \tag{35}$$

where $f(t_i)$ is the pollution stock of the country under consideration in year t_i, $N(t_i) \in \{0, 1\}$ indicates a rare disaster in year i, and $Z(t_i)$ is recovery if $N(t_i) = 1$[8]. As far as the the recovery distribution (3) for rare climate events is concerned, we assume that it follows a power law characterized by a parameter of $\beta = 6.27$ as estimated in [8] based on a large sample of rare disasters. The damage function D and rare-disaster frequency ν depend only on pollution stock value $\overline{f}$, so the expected value and variance of $\eta(t_{i+1})$

[8]We assume that the occurrence of two or more rare disasters in a single year is a low-probability event which can be neglected during the calibration of the model parameters.